Assignment 2

(Questions from Module 3 & 4)

Name 1	
Matric No.	
Section	
Date	

Name 2	
Matric No.	
Section	
Date	

Total Marks: 40

Instructions

- Students must answer **ALL questions** with complete workings, including conversions, arithmetic, and floating-point operations.
- Hardware algorithms should show full iteration tables with proper binary notation and indicate overflow, normalisation, truncation, or rounding.
- Complete this assignment in **pairs** (**2 members ONLY**) and **submit handwritten** work in **PDF format** through eLearning within 7 days, as specified by your instructor.

Collaboration Requirements

Work together with your assigned partner and ensure both students contribute equally to the assignment. Communicate clearly and divide tasks fairly from the start.

Submission Format

Write neatly by hand in black or blue ink on standard A4 paper, then scan and convert to PDF. Include your names, student ID numbers, course details, instructor name, and submission date on the cover page.

Deadline and Submission

Confirm your specific 7-day deadline with your instructor. Submit only through the eLearning platform and keep a personal copy for your records.

QUESTION 1 (10 marks)

Based on **Program 1**, answer the following questions. Assume the starting address to allocate data for **myBytes** is 4000 0000h:

```
1   INCLUDE Irvine32.inc
2
3   .data
4           myBytes    BYTE     10, 20, 30
5           myWords    WORD     1000, 500, 0
6           myDwords   DWORD    50000, 10000, 0
7
8   .code
9   main PROC
10
11          MOV ECX, 0
12          MOV EAX, 0
13
14          MOV ESI, OFFSET myBytes   ;  (i)   ESI  =  ____________
15          MOV EDI, OFFSET myWords ;    (ii)  EDI  =  ____________
16          MOV CL, [ESI]             ;  (iii) CL   =  ____________
17          MOV BX, myBytes + 1       ;  (iv)  BX   =  ____________
18          ADD ESI, 100b
19          MOV EDX, (ESI)            ;  (v) EDX =  ____________
20
21  L1:
22          MOV  BX, myWords [EAX + 2] ;  (vi) BX  =  ____________
23          INC EAX
24          CALL dumpregs
25          LOOP L1
26  exit
27  main ENDP
28
29  END main
```

Program 1

a) Write the content of each register labeled as (i) - (vi) in **Program 1** as the instructions execute in sequence up to **line 22**. [6 m]

b) What are the addressing modes used in lines 14, 18, 19, and 22? [4 m]

QUESTION 2 (10 marks)

Based on **Program 2**, answer the following questions:

```
.data
   setArr1   WORD    105, 1011b, 6Ah
   setArr2   BYTE    15, ?, 2 DUP (90), 'CAR'

.code
main PROC
   MOV AX, setArr1+2       ; (i)   AX = ________
   MOV DH, setArr2+3       ; (ii)  DH = ________
   MOV BX, setArr1+3       ; (iii) BX = ________
   MOV DL, setArr2+4       ; (iv)  DL = ________
exit
main ENDP
```

Program 2

a) Assuming a Little-Endian architecture, allocate the values of **setArr1** and **setArr2** in memory starting at address 3000h.[4 m]

b) Based on the given program, write the destination register and its content (in hexadecimal) for each instruction line labeled as (i) to (iv).[4 m]

c) Based on the program, if the programmer omits the CALL Dumpregs instruction, would the program still assemble and run without error? Give a justification.[2 m]

QUESTION 3 (10 Marks)

Given an equation for Q is:

$$Q = ((A + B) * C - D) / (E + F * G)$$

Write the program for Q using the following address instructions and identify the total of instructions lines for each of them:

a) Three-address instructions [2m]
b) Two-address instructions [2m]
c) One-address instructions [3m]
d) No-address instructions [3m]

QUESTION 4 (10 Marks)

Given an arithmetic equation for M is:

$$M = ((1 + 9) * 3) / (4 - 2) + 3$$

a) Construct the expression tree for the given arithmetic equation M.[2 m]

b) Convert the equation M into Reverse Polish Notation (RPN) postfix format. Calculate the value of M. [2m]

c) Write the program for Q using one-address and no-address instructions.
   i) One-address instructions [3m]
   ii) No-address instructions [3m]